

NISHANT CHAUDHARY

nishantchaudhary0512@gmail.com | LinkedIn | Portfolio | GitHub | +1 (930)-904-4657 | Bloomington, IN

EDUCATION

Indiana University Bloomington *MS, Computer Science*

August 2025 - May 2027 | **GPA** 3.76/4.0

Indian Institute of Information Technology, Pune *BTech, Computer Science*

August 2020 - May 2024 | **GPA** 8.04/10

WORK EXPERIENCE

ORIX | Azure AI Search, Document Intelligence, Databricks, Python, RAG, LLMs

AI Developer Intern

May 2026 - Present

- Architected an event-driven **Azure ingestion pipeline** handling **300-400 daily documents** into an **85,000+ document** knowledge base via **AI Document Intelligence**, **MarkItDown**, and semantic chunking.
- Engineered a **Retrieval-Augmented Generation (RAG)** agent over that corpus with **hybrid vector and keyword retrieval**, **reranking**, and prompt orchestration, replacing manual search for the operations team.
- Partnered with loan-portfolio analysts to deliver a **natural-language analytics layer on Azure Databricks** that compiles plain-English questions into **SQL** over millions of rows, taking query authoring from **10 minutes to under 20 seconds**.

NOMURA | Java, Spring Boot, React.js, Node.js, Python, Neo4j, Docker, Kubernetes

Software Engineer

July 2024 - July 2025

- Owned the **Angular to React migration** of a **40+ page enterprise portal**, re-architecting state management and caching to cut page load time by **84%** for **200+ daily users** across global desks.
- Built **Java Spring Boot microservices** and **REST APIs** for Nomura's internal developer platform, delivering an **auto onboarding service** that reduced application setup from **3 weeks to 5 mins**, saving **100+ engineering hours per team**.
- Modeled **100+ production applications** and their dependencies as a **Neo4j graph**, then optimized graph traversal and Python batch processing to reduce **API latency by 20%** at enterprise scale.
- Launched an **AI assistant** over the graph alongside a rebuilt **admin console** with usage analytics and monitoring, giving platform owners self-serve visibility across the application estate.

Software Developer Intern

January 2024 - June 2024

- Authored **unit and integration suites** for Java APIs and introduced **Pact Broker** contract testing across services, lifting coverage from **48% to 83%** and catching breaking changes before release.
- Rebuilt the portal as **20+ reusable components**, adopted as the shared design layer by teams serving **1,000+ users**.

TATA CONSULTANCY SERVICES | Python, PyTorch, OpenCV, FastAPI, React.js, Blender

Software Developer Intern

May 2023 - July 2023

- Trained a **medical imaging pipeline** over MRI/CT volumes spanning enhancement, **segmentation**, feature extraction, **tumor localization**, and stage classification, reaching **98.5% detection accuracy** on lesions down to **2-3 mm**.
- Reconstructed the nasal cavity in **Blender** across **22 anatomical regions** and served an **AR/VR particle-trajectory simulator** through **FastAPI** to a **React** interface in **under 15 seconds**, scoring drug deposition at predicted tumor sites **70% faster** than the synchronous build.

PROJECTS

Relay: LLM Cost Gateway  | Python, FastAPI, Redis Stack, PostgreSQL, Prometheus, Docker **July 2026**

- Built an **OpenAI-compatible LLM gateway** with **local-embedding semantic caching** (**95%** hit rate, **2.5%** false hits on an 80-pair holdout) and **complexity-based model routing**, cutting simulated API spend **91%** over **67k-request** load tests.
- Implemented **Lua token-bucket rate limiting**, per-team budgets, and **circuit-breaker failover** that held **zero dropped requests** through provider-outage drills at **3.6 ms p50** overhead, tracked in **Prometheus** and **Grafana**.

Argus: LLM Eval Platform  | Python, FastAPI, PostgreSQL, Streamlit, GitHub Actions, HDBSCAN **May 2026**

- Instrumented end-to-end tracing for **multi-step LLM chains** and auto-mined production traces into **eval datasets** using **HDBSCAN** clustering plus LLM labeling, adding **143 cases** at **92%** verified precision.
- Authored an **80-case adversarial benchmark** with a **six-flag safety judge** and paired **McNemar** CI gate that flagged every injected prompt regression before merge, backed by a **root-cause analyzer** at **81%** step-identification accuracy.

Cue: AI Workspace Agent  | Gemini, React, MongoDB Atlas, Model Context Protocol **February 2026**

- Created a **Chrome extension** pairing **Gemini** with **MongoDB Atlas** to forecast a user's **next five tasks** from browsing history, surfaced through a real-time **React** dashboard.
- Integrated the **Model Context Protocol (MCP)** to orchestrate actions across **9+ Google Workspace tools** and added a **multimodal scribe** producing meeting summaries and highlight reels.

SKILLS

- Languages & Frameworks:** Python, Java, C++, TypeScript, JavaScript, SQL, Scala, Spring Boot, FastAPI, React.js, Node.js, Angular, Flutter, Flask
- AI & Data:** PyTorch, TensorFlow, scikit-learn, HDBSCAN, RAG, Vector Search, PostgreSQL, pgvector, Redis Stack, RedisVL, Neo4j, Prompt Engineering, LLM Evaluation, CrewAI
- Cloud & DevOps:** Azure, Azure Databricks, GCP, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, Ansible
- Tools & Platforms:** MongoDB, MySQL, SQLAlchemy, Alembic, Streamlit, Docker Compose, Prometheus, Grafana, OpenTelemetry, Observability, Distributed Systems, MCP, Pact Broker